26900255 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEFSC301 - DNA ANALYSIS

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One Word Answer Question

Answer all questions

Each question carries **1** marks

1. What is the role of RNA polymerase? [U] / [CO1]
2. Which biological fluid is most commonly encountered as a source of nuclear DNA in forensic investigations? [A] / [CO2]
3. Which property of DNA allows individuals to be differentiated genetically? [E] / [CO2]
4. What is an ethical concern? [U] / [CO3]
5. State one forensic importance of population genetics. [U] / [CO1]
6. Which index is obtained by multiplying the paternity indices obtained from individual genetic loci? [E] / [CO2]
7. Which condition should be avoided during storage of biological samples to prevent DNA degradation? [E] / [CO2]
8. Which PCR stage separates the two strands of template DNA? [A] / [CO2]

9. Which value generally indicates stronger support for biological paternity: a higher or lower paternity index? [A] / [CO2]

10. What is a genetic marker? [U] / [CO4]

[1x10 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **3** marks

11. Apply DNA profiling to establish the identity of a person using recovered skeletal remains in a missing-person case [A] / [CO2]

12. Explain the inheritance pattern of Y-STR markers. [An] / [CO2]

13. Explain the role of positive and negative controls. [K] / [CO2]

14. Explain the role of expert testimony in DNA evidence. [An] / [CO3]

15. Discuss the importance of DNA discoveries in the development of forensic science. [K] / [CO1]

16. Discuss the usefulness of mitochondrial DNA in degraded biological samples. [U] / [CO2]

17. Explain the importance of traceability in DNA analysis. [K] / [CO2]

[3x5 = 15]

Part C

Short Essay Questions

Answer any **5** questions

Each question carries **6** marks

18. Apply the maternity index to determine the biological relationship between a mother and child in a disputed maternity case. [A] / [CO2]

19. Analyse the major limitations of DNA profiling. [An] / [CO4]

20. Explain the determination of allele frequencies using population data. [An] / [CO2]

21. Discuss the importance of human genetics in forensic science and biological identification. [U] / [CO1]

22. Describe the role of STRs in DNA profiling and their importance in forensic identification. [K] / [CO2]

23. Define DNA and explain its chemical composition and double-helical structure. [U] / [CO1]

24. Evaluate the suitability of hair as a source of DNA evidence and discuss the factors affecting DNA recovery. [E] / [CO2]

[6x5 = 30]

Part D

Essay questions

Answer any **1** questions

Each question carries **15** marks

25. Define DNA variation and explain polymorphism, single nucleotide polymorphism and their importance in genetic analysis. [U] / [CO1]

26. Evaluate the historical development of DNA profiling and critically discuss its importance and applications in modern forensic science [E] / [CO2]

[15x1 = 15]